Hints: Linear algebra – 9. Matrices over $\mathbb{Z}$ and over a finite field

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Existence: integer row and column operations (swaps, sign changes, adding an integer multiple of one row or column to another) are multiplications by unimodular matrices. Move a non-zero entry of smallest absolute value to position $(1,1)$ and divide with remainder by it along the first row and column. A non-zero remainder is a smaller entry, so restart with it. The process ends with d_1 in the corner and zeros in the rest of the first row and column.

Hint 2. To get $d_1 \mid d_2 \mid \ldots$: if some entry a is not divisible by d_1, add its row to the first row and repeat the reduction. This makes the corner entry strictly smaller (it becomes $\gcd(d_1, a)$). Then induct on the remaining block.

Hint 3. Uniqueness: by Cauchy–Binet every $k \times k$ minor of UAV is an integer combination of $k \times k$ minors of A. Since U, V are invertible over $\mathbb{Z}$, the gcd Δ_k of the $k \times k$ minors is invariant. For a diagonal matrix with $d_1 \mid \cdots \mid d_r$ it equals $d_1 \cdots d_k$, so $d_k = \Delta_k / \Delta_{k-1}$.

Hint 4. Applications: a finitely generated abelian group is $\mathbb{Z}^n / A\mathbb{Z}^m$ for an integer relation matrix A. Changing bases in $\mathbb{Z}^n$ and $\mathbb{Z}^m$ replaces A by its Smith form, so the group is $\bigoplus \mathbb{Z}/d_i \oplus \mathbb{Z}^{n-r}$. For $\Lambda = A\mathbb{Z}^n \subseteq \mathbb{Z}^n$ this gives $\mathbb{Z}^n/\Lambda \cong \bigoplus_i \mathbb{Z}/d_i$, of order $d_1 \cdots d_n = |\det A|$.

Problem 2

Hint 1. Induct on k, so only $k = 2$ is needed: $\det A = b_1 b_2$, and you want $A = B_1 B_2$ with $\det B_1 = b_1$. Use the Smith normal form $A = UDV$ to reduce to a diagonal D. Fix a sign by multiplying U by $\mathrm{diag}(-1, 1, \ldots, 1)$ if necessary.

Hint 2. For $D = \mathrm{diag}(d_1, \ldots, d_n)$ look for diagonal factors: $B_1 = \mathrm{diag}(e_i)$, $B_2 = \mathrm{diag}(f_i)$ with $e_i f_i = d_i$ and $\prod e_i = \pm b_1$.

Hint 3. If $\det A \neq 0$, then $b_1 \mid d_1 \cdots d_n$. Choose greedily $e_i = \gcd\bigl(d_i,\ b_1/(e_1 \cdots e_{i-1})\bigr)$; comparing prime exponents shows $\prod e_i = \pm b_1$. If $\det A = 0$, then $d_n = 0$. When $b_1 = 0$, take $B_1 = \mathrm{diag}(d_1, \ldots, d_{n-1}, 0)$ and $B_2 = \mathrm{diag}(1, \ldots, 1, b_2)$. The case $b_2 = 0$ is symmetric.

Problem 3

Hint 1. Everything is modulo 2. The determinant of $(a_{ij})_{i,j \in S}$ is the sum, over the permutations of S, of the corresponding products. Encode $A \bmod 2$ by the digraph with an arc $i \to j$ $(i \neq j)$ whenever a_{ij} is odd.

Hint 2. Claim: A is very odd if and only if all a_{ii} are odd and the digraph has no directed cycle. If there is a cycle, take a shortest one on the vertex set S. It has no chords, so only the identity and the cycle itself contribute to the determinant of the submatrix on S, which is $1 + 1 \equiv 0$.

Hint 3. Conversely, an acyclic digraph has a topological order. After a simultaneous permutation of rows and columns, $A \bmod 2$ is upper triangular with ones on the diagonal, and so is every principal submatrix. Powers of such a matrix are again upper unitriangular in the same order.

Problem 4

Hint 1. Two facts over $\mathbb{F}_2$: $x^T M x = 0$ for every symmetric M with zero diagonal, and $x^T x = \mathbf{1}^T x$ for every x, where $\mathbf{1}$ is the all-ones vector. Odd powers A^{2l+1} are again symmetric with zero diagonal, since $(A^{2l+1})_{ii} = (A^l e_i)^T A (A^l e_i)$.

Hint 2. Suppose $A^n e = \mathbf{1}$ for a basis vector e, and put $f(t) = e^T A^t e$. Then $f(\text{odd}) = 0$ and $f(n) = 1$. Show that $f(n+j) = f(2j)$ for all $j \geqslant 0$: compute $e^T A^j \mathbf{1}$ in two ways, using $\mathbf{1} = A^n e$ and $x^T \mathbf{1} = x^T x$ with $x = A^j e$.

Hint 3. Write $n = 2^s o$ with o odd and iterate $x \mapsto n + x/2$ starting at $x_0 = n$. The values $x_i = 2n - n/2^i$ satisfy $f(x_{i+1}) = f(x_i) = 1$ as long as x_i is even. Then $x_s = 2n - o$ is odd while $f(x_s) = 1$, a contradiction.

Problem 5

Hint 1. An odd square is $\equiv 1 \pmod 8$. Write the matrix as $J + 8B$, where J is the all-ones matrix and B is an integer matrix.

Hint 2. Expand $\det(J + 8B)$ by multilinearity in the columns. Every term with two or more columns of J vanishes, since $\operatorname{rank} J = 1$. Each remaining term has at least two columns from $8B$.